

Day-ahead electricity load forecasting based on hybrid model of EEMD and Bidirectional LSTM

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ABSTRACT

Load forecasting has always played a particularly important role in the power industry. In this article, we proposed a hybrid model based on Ensemble Empirical Mode Decomposition (EEMD) and Bidirectional Long Short-Term memory (Bi-LSTM). The original time series of load demand was decomposed into several sub-series called Intrinsic Mode Functions (IMFs) using the EEMD method. Then each IMF was divided into training, validation and testing datasets, and predicted using Bi-LSTM. Finally, the forecasting of the load demand would be obtained by composing the predictions of all IMFs. This model was applied to forecast the electrical demand in Hanoi using the data in 2018. The data was first decomposed based on Seasonal-Trend decomposition using the LOESS (STL) method to find down if there was a seasonal characteristic or not. Then 2 cases were analyzed, forecasting the load with and without considering the seasons. The performance of the proposed model in both cases were compared with other traditional models. The results showed that the proposed approach outperformed the other methods with minimal mean absolute percentage error (MAPE) of just over 2% when considering the whole year data as one time series. Meanwhile, the results of the seasonal division are slightly better in 3 seasons and more biased in the summer due to more variability.

KEYWORDS

Day-ahead load forecasting, hybrid model, Ensemble Empirical Mode Decompositions, Bidirectional Long Short-Term Memory

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1 INTRODUCTION

Ensuring the balance between electricity consumption and generation is critical for the reliable and effective operation of the whole electric power system. The load demand, however, strongly varies through the day and changes by season. Thus, in order to enhance the security and stability of the system, it is necessary for forecasting load accurately [1, 21]. An under-estimation of the load will affect the reliability of supplying the load and reduce the stability of the entire system, meanwhile, an over-estimation may increase the energy loss and operational costs [21]. There are various factors that affect the electricity profile, including energy policy, industrial manufacturing activities, the population, weather conditions, working days or holidays, and others [26]. Therefore, electric demand patterns exhibit non-linear, nonstationary, cyclic, seasonal, and stochastic characteristics. These lead to challenges in developing an electric load forecasting model.

Numerous forecast models for electricity load have been proposed and developed. The models can be classified as statistical and AI-based models [26]. Several studies used statistical models such as ARIMA [16, 18], the regression model [3], and the exponential smoothing model [22]. For example, an ARMA model with a multi-model partitioning filter to identify the ARMA parameters was used to predict the electricity load. The effectiveness of this model was proven by empirical results [16]. Rallapalli and Ghosh [18] proposed a multiplicative seasonal ARIMA (MSARIMA) method to predict monthly peak demand in India, and empirical results showed that MSARIMA outperformed other benchmark models. Although these models performed considerable accuracy, they assumed the time series is linear, thus they cannot achieve high accuracy when forecasting a non-linear time series [11]. Another approach for forecasting models is using AI-based methods. The AI-based models are gaining popularity due to their ability to model the nonlinear and complex patterns of the load [4]. Mansoor et al. compared 2 ANN-based load forecasting techniques and compared the accuracy of these 2 forecasting models through experiments [12]. In [8], the authors proposed a generalized regression neural network for estimation of electrical load forecast with the fruit fly optimization algorithm for training the network. A support vector regression-based (SVR) is proposed in [24], where sub-sampling is used to

speed up the SVR modeling process. Recently, due to the mature development of deep learning methods such as the long short-term memory (LSTM) technique, there was a trend in applying LSTM in the field of electric load forecasting. It has shown superior performance in comparison with traditional methods [13, 15, 19, 25]. The Bi-LSTM model has emerged in recent years thanks to the advantage of acquiring context information of time series data [2].

Moreover, in order to enhance the forecasting accuracy, hybrid or combined models have been commonly used. One of the approaches of the hybrid model is to decompose the historical load time series and combine it with any of the statistical/ML-based models [1]. Among of the decomposition methods, wavelet transform is a broadly used algorithm [17]. Ghofrani et al. [6] developed an ensemble electricity load forecasting model combining the Bayesian neural network with discrete wavelet transforms. A hybrid model of wavelet decomposition and genetic algorithm training optimization of ANN for day-ahead peak load was presented in [14]. Another common decomposition method is Empirical Mode Decomposition (EMD), which is a part of the Hilbert-Huang Transform (HHT). While wavelet transform decomposes the original time series into subseries by looking at the time frequency domain, EMD processes the time series data in the time domain [17]. Fan et al. [5] proposed an integrated model of SVR and differential empirical mode decomposition (EMD) and autoregression for load demand forecasting. The result indicated that the presented model can be applied to predict with good precision.

In this paper, we proposed a novel hybrid model based on ensemble EMD (EEMD) and bidirectional LSTM (Bi-LSTM) for short-term. The historical load data were decomposed using EEMD into intrinsic mode functions (IMFs), then these IMFs were predicted using Bi-LSTM models and finally the results were composed together for the entire forecasting of the load demand. Additionally, the seasonal characteristics of the load demand were also considered by using 4 separated models for 4 seasons and compared with 1 model for all seasons. The Seasonal-Trend decomposition using LOESS (STL) was applied to detect the trend and seasonal characteristics of the load time series.

The rest of this paper is organized as follows. Section 2 provides the basic methodologies including the decomposition techniques of STL, EMD, and EEMD, the forecasting models of LSTM and Bi-LSTM, as well as the proposed hybrid modeling framework. The prediction results and discussions of the proposed modeling framework were reported in Section 3. The original data set is divided into 2 parts (train and test) with the ratio of 80:20 and the forecast model is based on multi-step forecasting and used to forecast for the next 24 hours (day-ahead). Finally, the conclusions of this study are described in section 4.

2 METHODOLOGY

2.1 Decomposition technique

In this article, we introduce two decomposition methods. The STL method is to detect the trend and seasonal, and the EEMD method is to remove the noise and to make the forecast.

2.1.1 Seasonal and Trend decomposition using Loess (STL). As mentioned above, time series data have some complicated underlying

characteristics. To separate time series data into independent sub-components including Trend, Seasonal, Irregular remainder, we can use several decompose techniques. There are two main methods of decomposition: additive decomposition and multiplicative decomposition, which were written with

$Y_t = T_t + S_t + R_t$ and $Y_t = T_t \cdot S_t \cdot R_t$ respectively, where Y_t is the dataset, T_t is the Trend component, S_t is the Seasonality component, R_t is the Residual component.

In this paper, we applied STL method which is an additive decomposition method and is applicable to any dataset. However, meaningful results are returned only if a periodic time pattern exists in the data. There are two recursive procedures of STL decomposition: an inner loop and an outer loop. The inner loop fits the trend and estimates the seasonal component. Every inner loop consists of six steps in total:

1. Calculate detrended series.
2. Sub-cycle smoothing. To smooth the subgroups of values at each position of the season cycle we use LOESS
3. Low-Pass Filtering of Smoothed Cycle-Subseries. This procedure consists of two MA filters and a LOESS smoother.
4. Detrending of Smoothed Cycle-Subseries.
5. Eliminate seasonality.
6. Using LOESS to smooth the series has removed seasonality to capture trends element of this card.

2.1.2 Ensemble Empirical Mode Decomposition (EEMD). Empirical Mode Decomposition (EMD) method was proposed by Huang et al. 1998 [9]. The meaning of IMF is the mono component function or an oscillatory mode with one instantaneous frequency. Any non-linear and non-stationary data set can be adaptively decomposed into IMF components by this method. In the EMD approach, the data $X(t)$ was decomposed into terms of Intrinsic Mode Functions (IMFs). The decomposed IMF features are linear and stationary.

$$X(t) = \sum_{j=1}^n C_j(n) + R_n \quad (1)$$

where $C_j(n)$ is the i^{th} IMF component, R_n is the residual of the data after n number of IMFs were decomposed. Each IMFs contains different amplitude and frequency components, but they have the same following characteristics:

- The IMF strings have the equal number of extrema and zero-crossing. These numbers can be different from the original data set.
- At any data point, the mean value of the envelope is determined by the local maxima, and the envelope defined by the minima is zero.

However, the major shortcoming of EMD is the effect of mode mixing. This drawback is expressed when the oscillations with disparate time series are preserved in one IMF, or the oscillations with the same time series are split into different IMFs.

To solve this problem, a new method named “Ensemble empirical mode decomposition” has been proposed by Wu and Huang [23]. The idea was to add the white-noise series into the original data set before decomposing. White noise is added to the original signal and distributed evenly to all time-frequency space [20]. Each IMF is then predicted separately, and the forecasting results are averaged.

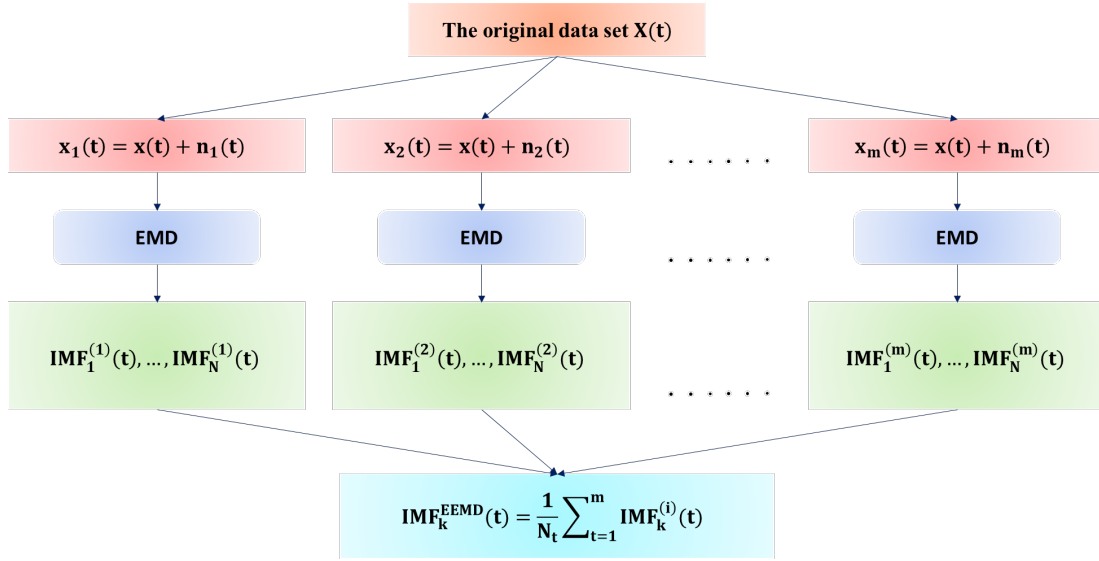


Figure 1: Diagram of the EEMD method

The EEMD block diagram is shown in Figure 1. The procedure of the EEMD process is explained as below [10]:

1. Add Gaussian White noise to the input signal $x(t)$ to create noise added input $x_1(t)$
2. Decompose $x_1(t)$ to different independent IMFs and one residue according to Eq. 1 using EMD method.
3. Repeat steps 1 and 2 using different noise each time
4. Average IMFs to get the final value.

2.2 Forecasting model

2.2.1 Long-short term memory. Long short-term memory (LSTM) [7] is an innovative application of cyclic neural networks (RNNs), which works so efficiently on a variety of problems and has gradually become as popular as they are today. LSTM networks were first proposed by Hochreiter and Schmidhuber in 1997. LSTM is one of the suitable deep learning architectures for time series prediction problems because these networks are very good at capturing the dependencies long term. To avoid vanishing gradient like in RNN, LSTM has three multiplicative units, which are input gate (i_t), forget gate (f_t) and output gate (o_t) are designed to help store information, write, or read from cells, and with the update gate ($\tilde{C}_t$) update the current state of the cell. Assuming we have input $X = x_1, x_2, x_3, \dots, x_t$ and output $Y = y_1, y_2, y_3, \dots, y_t$ with t showing the time, then we have to filter the information from x_t at time t of the input gate (Eq.2) and then create an update implementation to update status information (Eq.3)

$$\text{Input gate : } i_t = \sigma. (W_i. [h_{t-1}, x_t] + b_i) \quad (2)$$

$$\text{Update gate : } \tilde{C}_t = \tanh. (W_C. [h_{t-1}, x_t] + b_c) \quad (3)$$

Then, the forget gate filters input historical information, discard unimportant information, and pass on long-term trend information to $t + 1$. (Eq.4)

$$\text{Forget gate : } f_t = \sigma. (W_f. [h_{t-1}, x_t] + b_f) \quad (4)$$

By removing an unimportant piece of information from the history and adding valuable information to be extracted, the state of the old cell C_{t-1} is upgraded to the new state C_t (Eq.5)

$$\text{Cell gate : } C_t = f_t. C_{t-1} + i_t. \tilde{C}_t \quad (5)$$

Then, the output gate o_t and the hidden layer state h_t of the memory cell can be written as (Eq.6), (Eq.7), respectively:

$$o_t = \sigma. (W_o. [h_{t-1}, x_t] + b_o) \quad (6)$$

$$h_t = o_t. \tanh(C_t) \quad (7)$$

Finally, the output of the final forecasting result y_t (Eq.8)

$$y_t = W_y. h_t + b_y \quad (8)$$

In the above formulas, W_i, W_f, W_o, W_c are the weight matrix of each layer. While h_{t-1} and h_t denote the state of the hidden layer, C_{t-1} and C_t show the state of cell at $t-1$ and t time, separately. $0 < f_t, i_t, o_t < 1$; b_f, b_i, b_o are the bias of each gate and b_y is the output bias. $\sigma(\dots)$ and $\tanh(\dots)$ are the hyperbolic tangent function [27] and the sigmoid function [28], in that order.

The following equation is to determine the update gate.

$$u_t = \tanh. (W_C. [h_{t-1}, x_t] + b_c) \quad (9)$$

The forget gate decides the number of cells needed, and the function C_t is the way to determine the number of inputs, hidden layers of the previous state. This process is shown in Figure 2(a).

2.2.2 Bidirectional Long-short term memory networks. Bi-directional long short-term memory (Bi-LSTM) is a hybrid of Long Short-Term Memory (LSTM) and a Bi-directional Recurrent Networks (Bi-RNN). Bi-LSTM is a processing model consisting of two LSTMs: one that takes input in the forward direction and the other in the reverse direction. Bi-LSTM effectively increases the amount of information available to the network, improving the content available to the algorithm. The Structure of Bi-LSTM is illustrated in Figure 2(b).

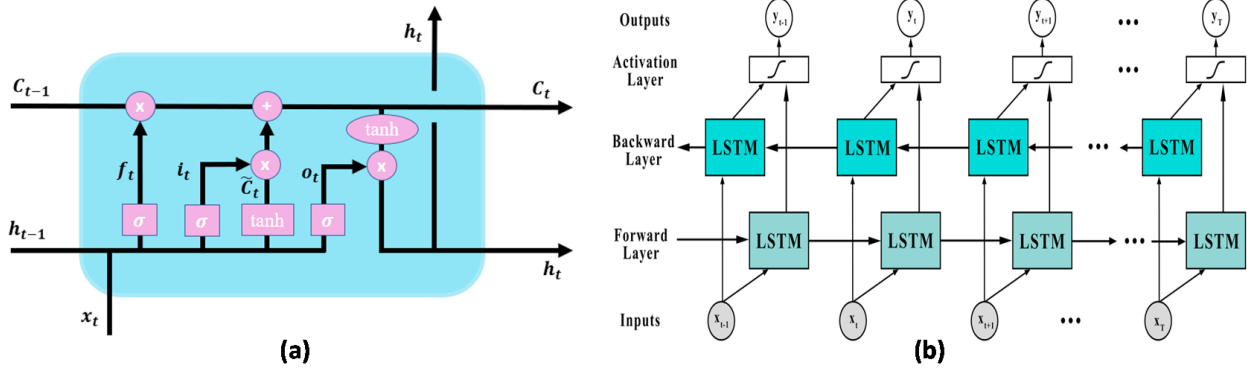


Figure 2: Structure of a LSTM cell (a) and Bi-LSTM network (b)

2.3 The proposed hybrid model

The main idea of the proposed model was to separate the power supply chain into sine-form components. Next, each IMF was analyzed with its delays. The IMFs had low correlations that were considered as noise, then we trained each Bi-LSTM model for the remaining IMFs, the residual component (noise) was removed. Finally, the results were synthesized to provide the final forecast for electrical loads. The details of the proposed model are described in Figure 3

2.4 Performance evaluations

To evaluate the predicted value derived using proposed method, the accuracy was evaluated according to Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE).

$$RMSE = \sqrt{\frac{\sum_{t=1}^n (Y_t - \hat{Y}_t)^2}{n}} \quad (10)$$

$$MAE = \frac{\sum_{t=1}^n |Y_t - \hat{Y}_t|}{n} \quad (11)$$

$$MAPE = \frac{1}{n} \sum_{t=1}^n \left| \frac{Y_t - \hat{Y}_t}{Y_t} \right| \quad (12)$$

where n is the number of observations, Y_t is the actual value, $\hat{Y}_t$ is the predicted value.

3 RESULTS AND DISCUSSIONS

3.1 Dataset information and analysis

The original dataset was the electrical load in Hanoi in 2018 collected by Vietnam Electricity Corporation (EVN), the overview of the datasets is shown in Figure 4. The electrical load data was sampled by hour and included 8685 data points corresponding to 1 year. In this paper, two study cases have been discussed, which are with and without seasonality.

3.2 Data preprocessing

3.2.1 Handling the missing data. During data collection, some data points had missing values. In this article, we used the proposed

forecast model to predict the missing data. Since the number of missing data was very small compared to the original data set, the errors might not affect too much to the accuracy of the forecast model.

3.2.2 Decomposition and denoising data. There have been many methods of decomposition and denoising data to be used in literature, but the authors often used their experience to identify the noise components, which may lead to the exact lack of the forecast model. We proposed to apply the EEMD method for decomposition and autocorrelation functions (ACF) to reduce the noise for the datasets. Low-correlation IMFs were considered to be the noise which was unpredictable and therefore could be removed.

Furthermore, the STL method was used to detect the trend and the seasonality of the datasets, thereby splitting the data string into 4 seasons to make the forecast more accurate. From Figure 5, we can see a clear difference in the electrical load chart over time. Therefore, dividing the time-series into 4 seasons would be able to improve the accuracy of the forecasting model.

3.3 Model parameters and processing time

For each IMFs, we build a suitable Bi-LSTM forecast model. For example, for the IMF2 string the appropriate parameters are selected and shown in the picture below:

According to Figure 6, to avoid over-fitting, a dropout layer is added with the rate of 0.1-0.2. The Bi-LSTM layer, which is considered the main component of the model is set to 256 filters. The epochs have to be big enough to avoid underfitting (epochs are set to 100 for the IMF2), but also not too big because it could affect the processing time and the learning performance of the model. Table 1. shows the synthesis of parameters of individual models. In this model, not only a single layer of Bi-LSTM has been used but also some LSTMs are added to increase accuracy. The hyperparameters are chosen based on the author's experience and could be adjusted to obtain optimal results for each IMF by training several times. In the more complicated model, the processing time would be longer, however, the accuracy could be enhanced. This research is working on an Intel(R) Xeon(R) CPU E3-1231 v3 @ 3.40GHz with 16.00GB RAM PC. For each IMF, the computer needs about 1.5-2 mins to

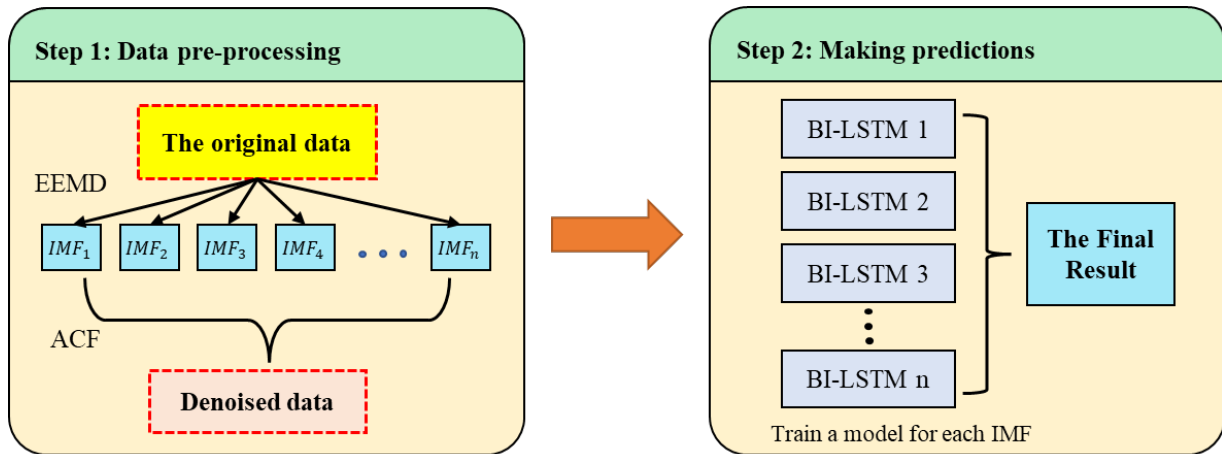


Figure 3: Construction of the proposed model for load forecasting

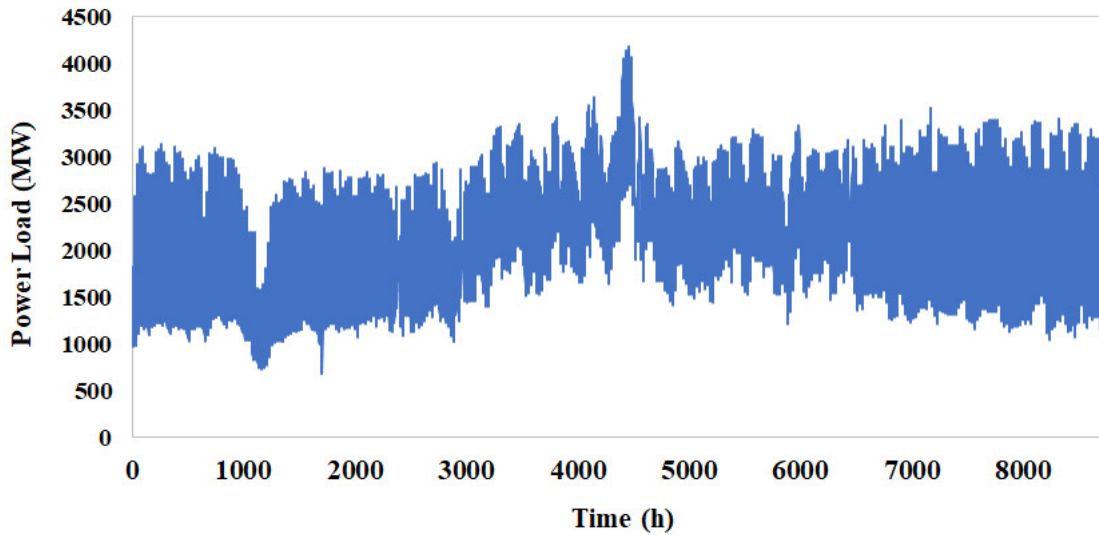


Figure 4: Full-year load data in Hanoi in 2018

solve an epoch, but only 30-40 minutes to give the final forecasting result thanks to the Keras Callback function.

3.4 Forecasting results and evaluations

As the STL result, we noticed the crop of the original dataset, so the forecasting is made on 5 datasets including the original data and 4 season datasets. Each dataset has its own characteristics.

3.4.1 Study case 1: Days-ahead forecasting based on the original dataset. In this case, the EMD, EEMD method were used to decompose and denoise the original load consumption data as shown in Figure 7a, 8a. 7b, 8b shows the correlation of IMFs, the first IMFs had a very low correlation, so IMF1 was the noise and should be removed. Accordingly, the forecasts were performed on the remaining series after removing the IMF1.

The summary result of the comparison models based on the original dataset were demonstrated in Table 2. The proposed model shows the best accuracy.

3.4.2 Study case 2: Days-ahead forecasting based on the seasonal dataset. The decomposition and processing methods were similarly to the first case; however, the time series was divided into 4 seasons based on the STL method.

Each season may have separate characteristics which may affect the forecast results. For example, in spring and winter, the chart was quite stable, the forecast results could perform better. On the other hand, in summer, the demand of people could change continuously, the chart has a clear variation, whereby the error would increase. Once again, the proposed model shows the effectiveness in all 4 data sets. However, the error of the summer data set has increased due to its unstable.

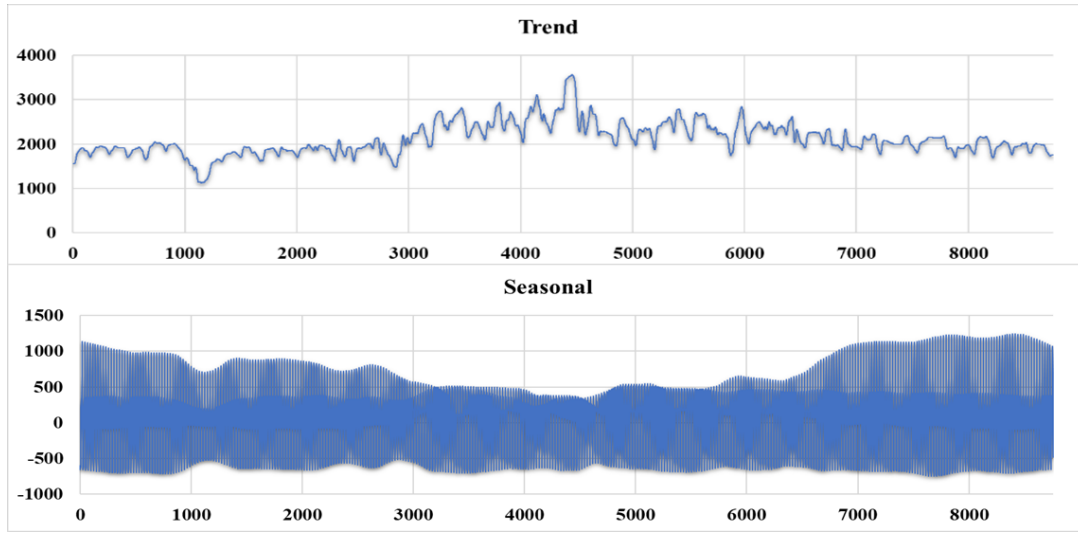


Figure 5: Time series decomposition using STL on the original dataset

The findings of Table 2-3 and Figure 9, 10, 11 for both study cases were summarized below:

1. It was observed that decomposition-based forecasting models outperform individual models in these cases.
2. Among the four decomposition-based hybrid models, the EEMD-Bi-LSTM model exhibits better performance with minimum MAPE, MAE, and RMSE values.
3. The EEMD-Bi-LSTM model achieves a superior prediction accuracy with a minimum MAPE value of 2.091972 and an RMSE value of 56.85947.
4. Compared to unidirectional LSTM models, the bidirectional LSTM model produces better prediction accuracy in terms of MAE, RMSE, and MAPE.
5. It was also observed that among the decomposition techniques employed in the study, hybrid models pre-processed using EEMD have superior performance compared to the EMD decomposition technique.
6. In different experiments, the error values are not in the same range. The MAPE values vary in the range of 2.01% to 6.41% and 0.42% to 6.68% for the original and the seasonal datasets, respectively. This may be due to the fact that the forecasting accuracy depends on the datasets employed and the prediction models involved.
7. The forecasting accuracy of the hybrid Bi-LSTM model was better than the single Bi-LSTM model. For instance, compared to the Bi-LSTM model, MAPE of the proposed model were ascended by 2.09% and 3.12%, respectively, for the spring dataset.

Through the above summary, it can be concluded that the proposed model provides the best performance, in all test cases.

4 CONCLUSIONS

In this study, a decomposition-based load forecasting model using bidirectional LSTM network was proposed. The load data was pre-processed and decomposed into IMFs by EEMD. Then Bi-LSTM

Model: "sequential_1"

Layer (type)	Output Shape	Param #
lstm_4 (LSTM)	(32, 30, 100)	40800
dropout_1 (Dropout)	(32, 30, 100)	0
lstm_5 (LSTM)	(32, 30, 100)	80400
lstm_6 (LSTM)	(32, 30, 100)	80400
bidirectional_1 (Bidirection	(32, 256)	234496
dense_2 (Dense)	(32, 100)	25700
dense_3 (Dense)	(32, 1)	101
Total params: 461,897		
Trainable params: 461,897		
Non-trainable params: 0		

Figure 6: The parameters of the IMF 2

was applied to train and forecast each IMF. Three criteria, namely MAPE, RMSE, and MAE were used to evaluate the forecasting accuracy. The proposed model obtained the best accuracy when compared with other methods with the error of over 2%, 61MW, and approximately 48MW, respectively. Additionally, the season of the load was also considered. The data was divided into 4 times series according to 4 seasons and applied the proposed method to predict. In general, the results of the proposed model still appear to be by far the best over the other methods. The comparison of two cases, without and with consideration of the seasons were conducted. From an overall perspective, the results of the latter were better than that of the former in Spring, Autumn, and Winter. Nevertheless, in Summer, the results of case two seem to be relatively high. One of the reasons was that the load in Hanoi in the summer fluctuated when the demand for electricity increased leading to a certain unpredictability.

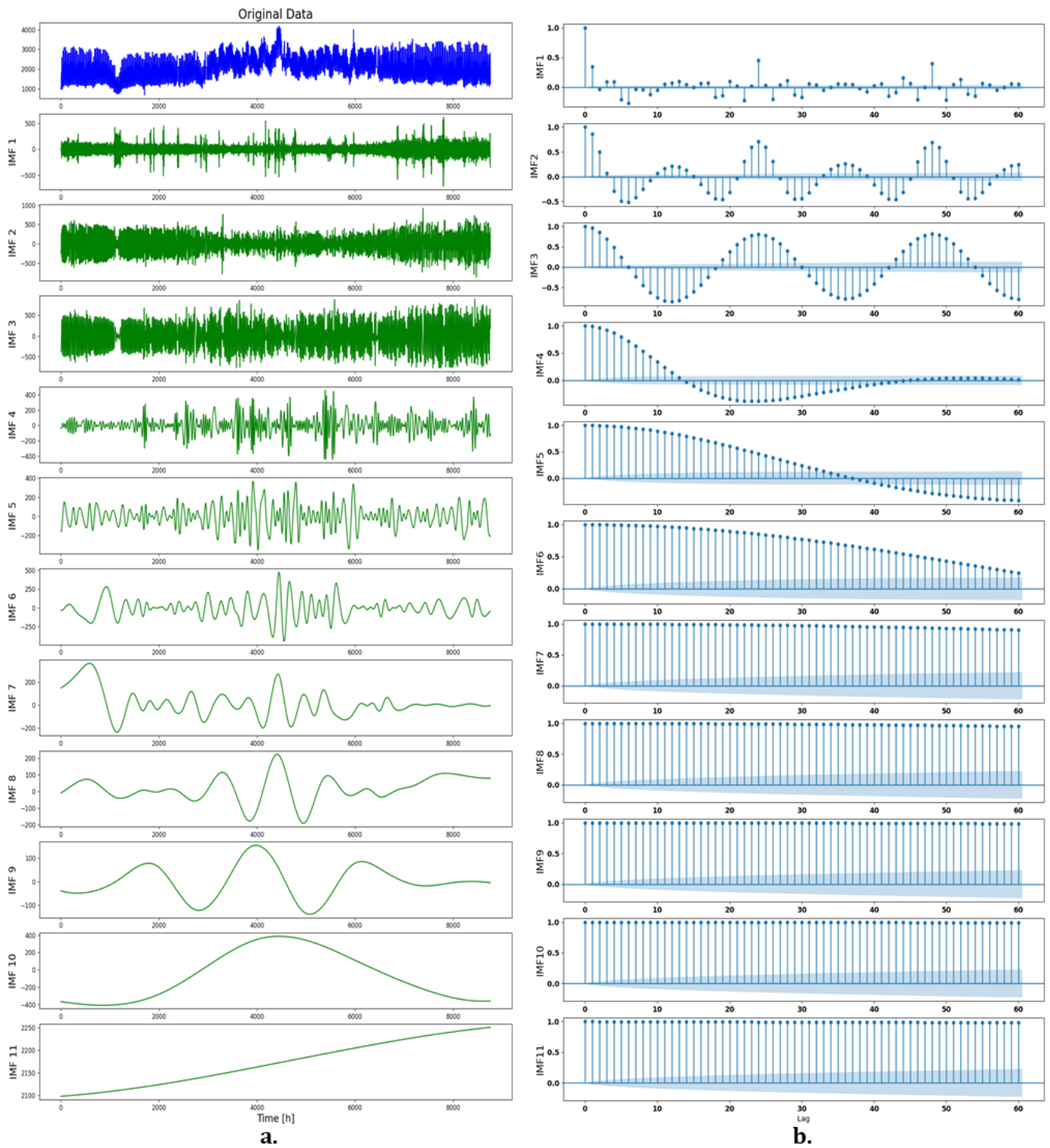


Figure 7: Time series decomposition using EMD on the original data set (a) and the ACF of the IMFs (b)

Table 1: Configuration of individual models

Model	Parameters	Number of Type
ARIMA	p, d, q orders	(8, 0, 1)
ANN	Number of hidden layers	4
	Number of neurons in hidden layers	(100, 50, 25, 100)
LSTM	Number of hidden layers	2
	Number of neurons in the LSTM layer	100
	Activation function of the output layer	ReLU
	Optimizer	Adam
Bi-LSTM	Number of hidden layers	3
	Number of neurons in the LSTM layer	100
	Number of neurons in the Bi-LSTM layer	256
	Activation function of the output layer	ReLU
	Optimizer	Adam

Table 2: Evaluate the accuracy of the models on the original dataset

Evaluation Criteria	ARMA	ANN	LSTM	EMD-LSTM	EEMD-LSTM	Bi-LSTM	Proposed model
MAPE	6.73160657	6.281504	8.960814773	6.613065756	5.645162369	4.590732743	2.558397
RMSE	189.961852	152.0939	216.3684382	178.6651624	179.0138923	132.6401207	61.05604
MAE	132.494059	118.263	153.7846582	125.6616205	118.6298682	88.28466244	47.97838

Table 3: Accuracy evaluation of the forecasting models on the seasonal datasets

Season	Evaluation Criteria	ARMA	ANN	LSTM	EMD-LSTM	EEMD-LSTM	Bi-LSTM	Proposed model
Spring	MAPE	6.124196	8.030966	8.680815	5.042563	4.683169	3.124982	2.091972
	RMSE	139.9742	191.4004	197.4435	125.814	123.8662	80.46658	56.85947
	MAE	109.6071	147.1115	161.4322	95.04482	88.46316	59.22196	40.21628
Summer	MAPE	4.235427	4.375871	9.537273	3.935133	4.662167	4.840948	3.019612
	RMSE	167.9214	214.0845	257.8151	155.552	212.2862	173.4219	121.5836
	MAE	110.3387	115.4684	191.0278	103.2272	123.9824	128.0371	78.90198
Autumn	MAPE	4.200932	4.0834	8.730091	3.901149	3.86235	2.750134	2.334691
	RMSE	126.0804	127.4886	206.3644	126.723	123.2849	74.2248	74.20301
	MAE	93.53556	93.0757	200.4849	90.3773	88.95446	58.32826	52.00852
Winter	MAPE	5.606503	4.630844	7.367269	5.492977	5.067227	4.877003	2.285433
	RMSE	135.8779	143.4229	183.2435	156.9906	136.9924	108.4927	57.28166
	MAE	109.2609	94.86371	139.1953	111.7856	99.8914	84.19567	43.45127

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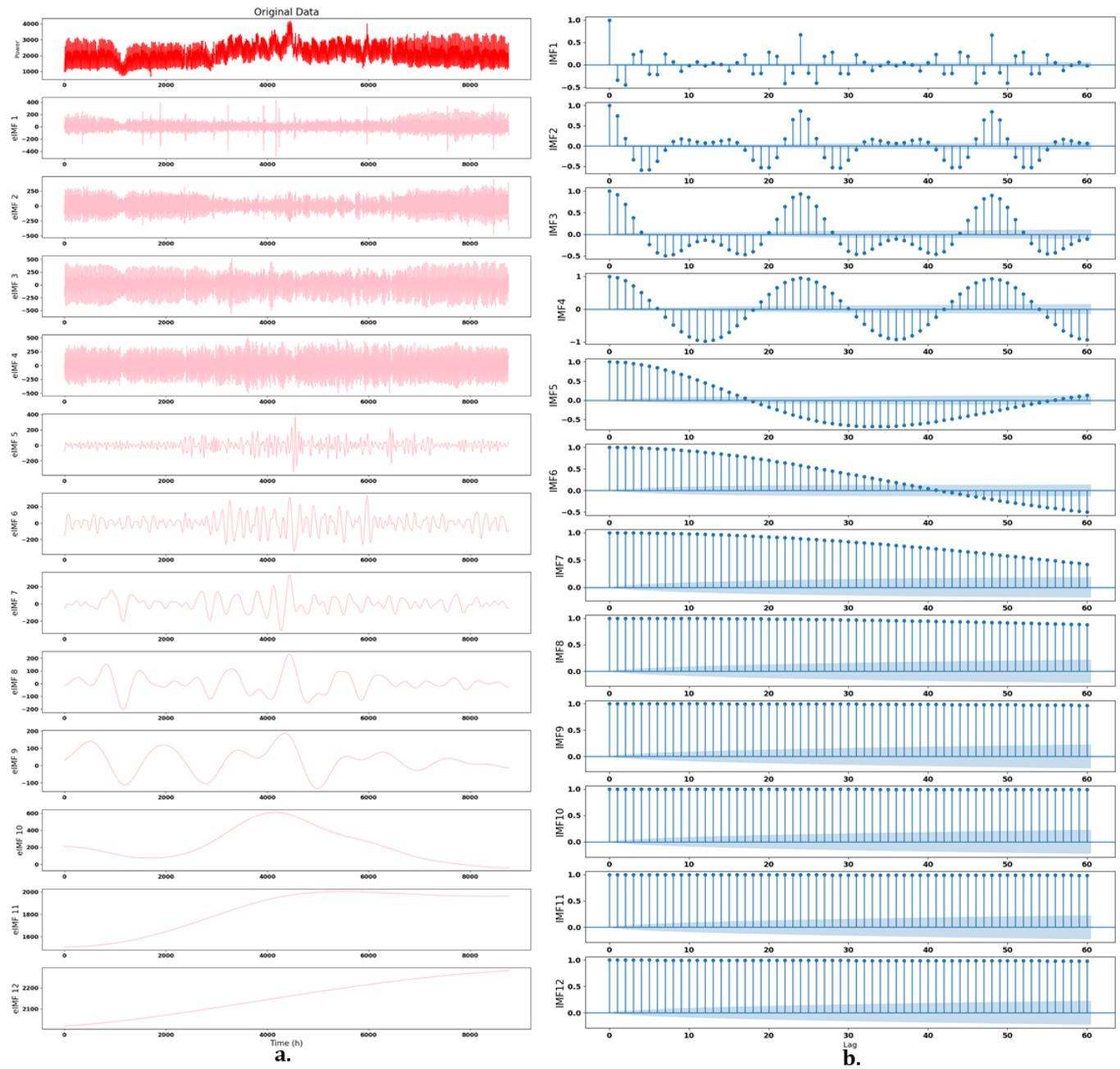


Figure 8: Time series decomposition using EEMD on the original data set (a) and the ACF of the IMFs (b)

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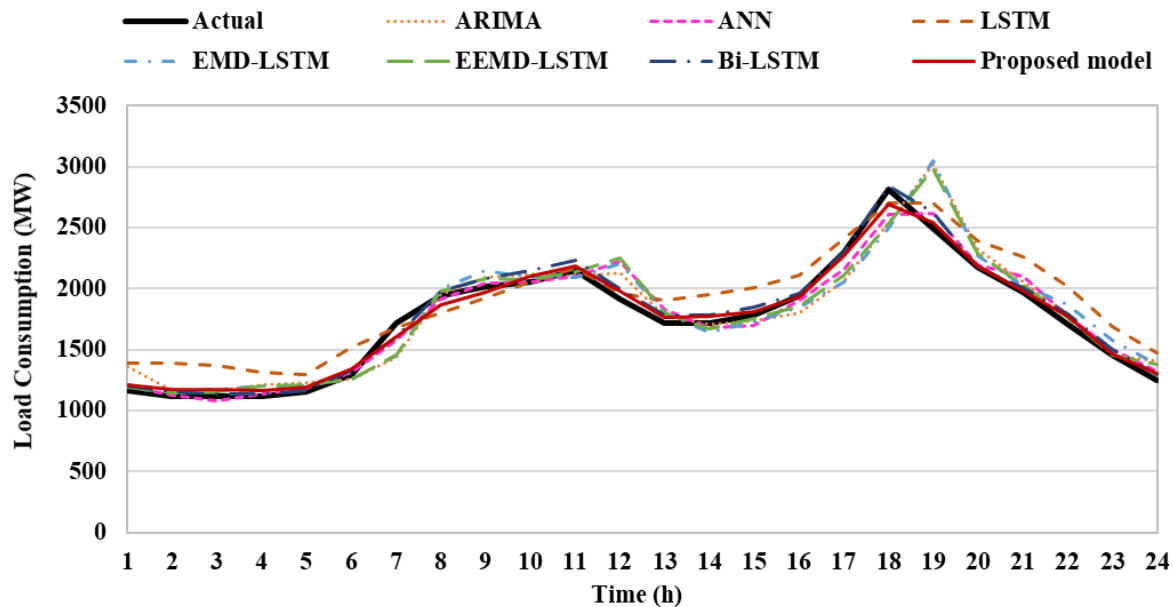


Figure 9: Result of day ahead load forecasting for 1 day.

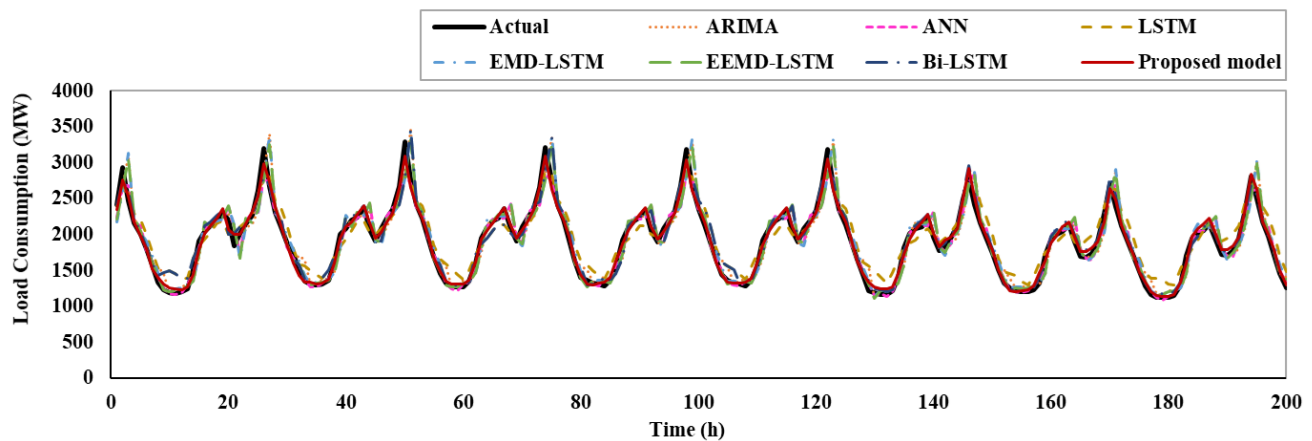


Figure 10: Forecasting results comparison based on the original dataset (200h)

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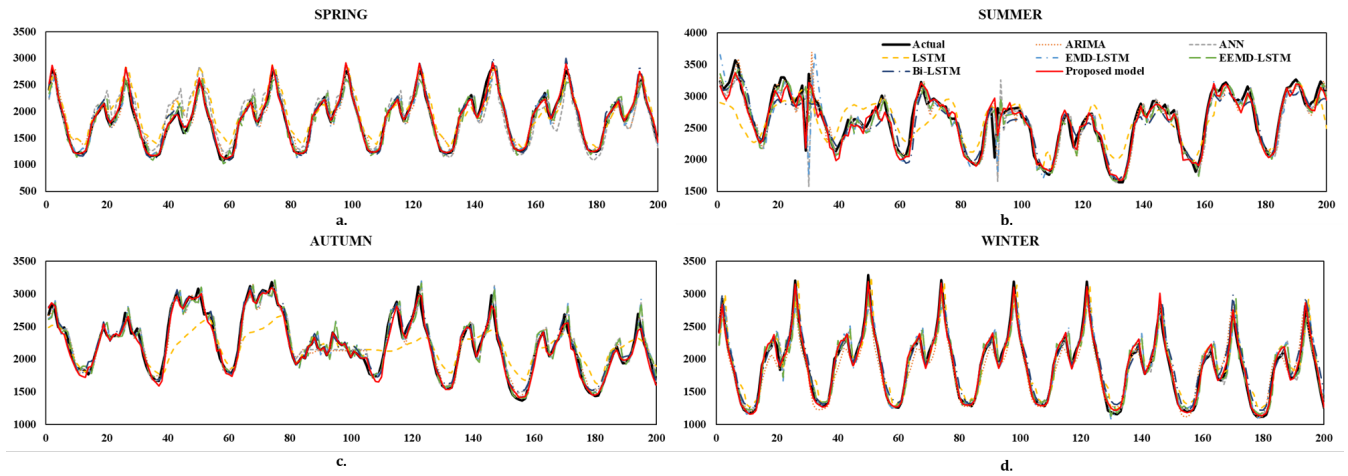


Figure 11: Forecasting results comparison based on seasonal datasets

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